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EFFORTS TOWARDS NEMO4 AND DEVELOPMENT PLANS FOR A CONTRIBUTION TO THE NEMO CONSORTIUM

F. Dupont and collaborators

MSC-CCMEP-CMDEM

MRD-RPNEM

Canada

NEMO 4

- Most options are now namelist parameters instead of CPP keys
- WAD
- Adaptive implicit vertical advection
- New HPG
- Wave-ocean interface
- Sub-tiling
- ABL
- Shallow water mode
- Mixed precision via a reduced precision emulator library for Fortran

NEMO 5

- RK3 => larger time-step
- Reduction of MPI communications (i.e., larger halo zones and mix OMP-MPI?)
- GPU-capabilities with Psyclone
- Missing z tilde
- LIM3 updates (form drag, landfast ice mask, gravity drainage)
- 5-band light penetration
- BCG updates
- Geometric lateral closure schemes
- AGRIF updates

Being a member of the NEMO consortium

- The consortium has been ready to accept us since 2019.
- It will ensure that Canadian model strategy is considered in consortium developments.
- Will increase our knowledge of the code and of its developments.
- Better integration of Canadian code into NEMO.
- ensure better alignment with other NEMO international partners.
- Will require a dedicated resource (1 FTE)

Canadian (potential) additions to the code

- Special SSH OBC
- Extending WAD
- Absolute vertical velocity
- Some extensions to tides
- AGRIF (future developements, Paul?)
- Tools (OBC, runoff, evaluation)
- Spectral nudging
- Need to address GPU (NEMO5 only)
- NEMO-CICE interface
- Wave-coupling interest
- Improving the BBL
- NH interest but no chance to get it in NEMO

CONCLUSIONS

- Switch to CICE6 and NEMO4 was painful and error-prone due to the large differences (icepack, defaults, new loop system in NEMO…etc)
- Maintenance should be easier now. If we could get our extensions in the base code, it would be even better! -> consortium member!
- We control now the CICE driver and to a lesser degree the NEMO-CICE interface
- Icepack is used as the internal engine of many sea-ice models (the one in MOM, and FESOM), could be also used in SI3? although performance issues may be a limiting factor…
- Comparison for CREG025+CORE2 configuration was good.
- Who is working with NEMO5?
- Who wants to contribute?

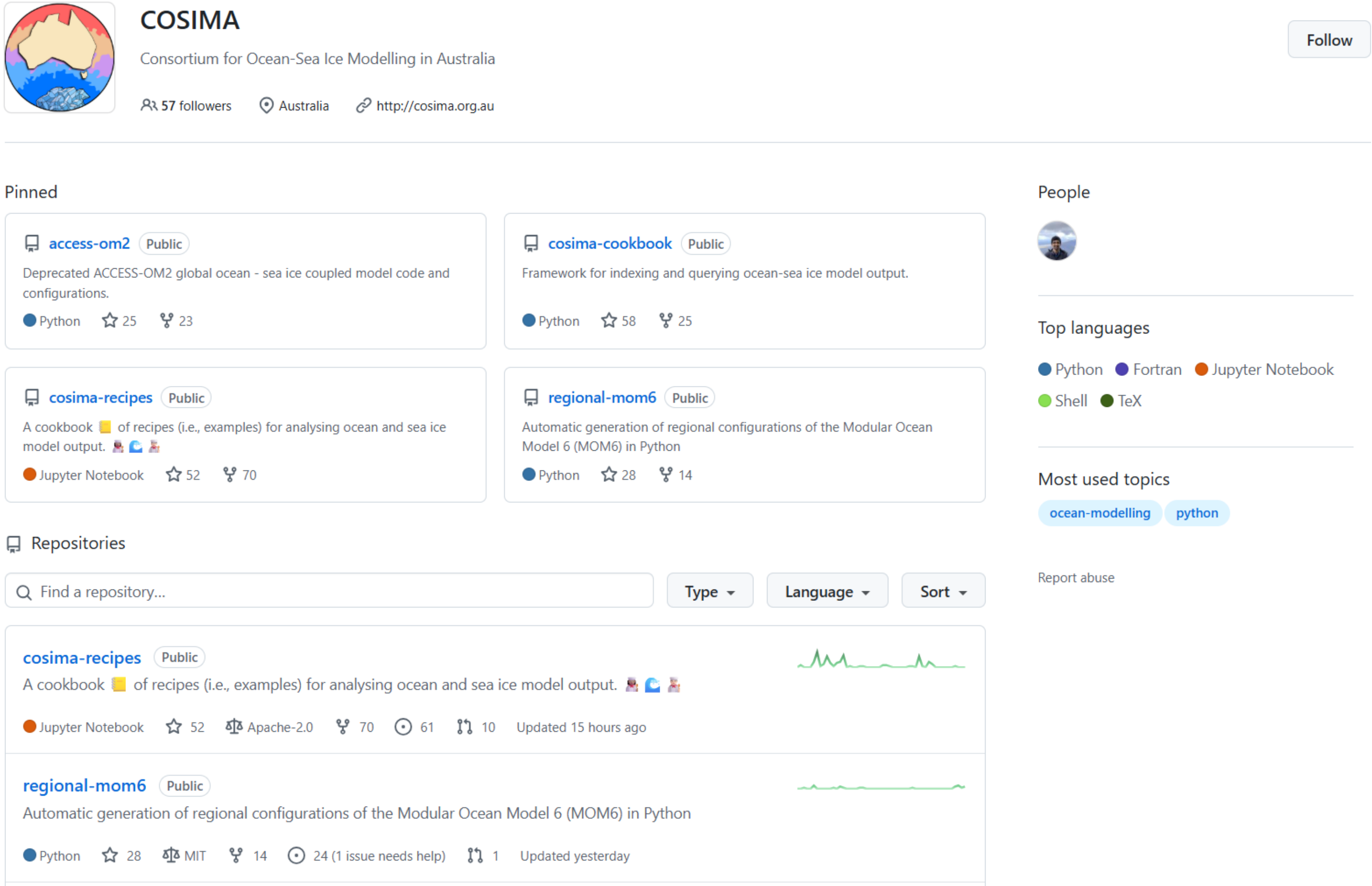
Cross institute coordination of NEMO development

Natasha Ridenour

Discussion points

- What is the end goal/vision?
- Current barriers that could be removed?
- Communication
  - NEMO seminars
  - Online forum?
  - Github/gitlab
  - Webpage that has all the people/models/work, talk links/channels etc?
    - It exists…

Examples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Public
A cookbook of recipes (i.e., examples) for analysing ocean and sea ice model output.
Jupyter Notebook 52 70
regional-mom6
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Automatic generation of regional configurations of the Modular Ocean Model 6 (MOM6) in Python
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